[INCH-POUND]
A-A-3021
October 10, 1996

COMMERCIAL ITEM DESCRIPTION

EXTRACTOR, SCREW (STUD AND PIPE)

The General Services Administration has authorized the use of this Commercial Item Description for procurement of type I, class 1, style A and B; class 2; and class 3 extractors specified in GGG-E-936C for all federal agencies.

1. SCOPE.

1.1 Scope. This Commercial Item Description covers hand-operated extractors for use in removing broken screws, studs, bolts, and pipes.

2. CLASSIFICATION.

2.1 Classification. This Commercial Item Description covers screw (stud and pipe) extractors of the following classes and styles:

Class 1	-	Tapered, spiral flutes or ridges
Style A	-	Drill
Style B	-	Plug
Class 2	-	Tapered, straight fluted
Class 3	-	Straight, straight fluted

3. SALIENT CHARACTERISTICS.

3.1 Design. The screw (stud and pipe) extractors shall be of steel and shall be the standard product of the manufacturer. A square end or other suitable means shall be provided for using a standard wrench in backing out broken screws, studs, bolts, and pipes. Dimensions not having a minimum, maximum, or tolerance range are considered to be nominal dimensions. Size deviations that are normal to standard manufacturing practices will be acceptable. The recommended drill size required for using the screw extractor shall be permanently and legibly marked on each extractor; the nominal pipe size shall be permanently and legibly marked on the class 3 screw extractors. When required to be tested for hardness, sample extractors shall be tested in accordance with ASTM E 18. When required to be tested for torsional moment, sample extractors shall be tested as follows:

a. Class 1, style A, sizes 7 through 12, inclusive
Class 1, style B, sizes 6 through 9, inclusive

A hole shall be drilled in a steel plate or block, the hole being determined by referring to the largest drill size for the specific class, style, and size of the extractor listed in the applicable table in this commercial item description. The steel plate or block shall be of carbon tool steel and heat treated to a hardness of 36 to 42 on the Rockwell C scale.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center, Kansas City, MO 64131.

Distribution Statement A. Approved for public release; distribution is unlimited.

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b. Class 1, style A, sizes 4 through 6, inclusive
Class 1, style B, sizes 2 through 5, inclusive, and including size 10
Class 2, sizes 4 through 10, inclusive

The hole size shall be determined as in "a" above. The steel plate or block shall be of medium carbon steel and heat treated to a hardness of 24 to 30 on the Rockwell C scale.

c. Extractors not included in "a" and "b" above:

The hole size shall be determined as in "a" above. The steel plate shall be of low-carbon, unhardened steel.

If lead holes are not drilled, care shall be exercised to select drills that are ground on center, in order to prevent the holes from being drilled oversize. A new hole is required for each extractor when testing classes 1 and 2. The extractor shall be inserted in the drilled hole and gently driven in sufficiently to provide a firm engagement. A torque testing machine, or other suitable means, shall be utilized to exert a steady pressure of torque until the minimum torsional moment in inch-pounds is reached, or the extractor fails. Failure shall be deemed to have occurred when the extractor breaks or a permanent bend or twist occurs. If more than one of the sample extractors fail to meet the minimum inch-pounds specified in Tables I, II, III, or IV, as applicable, this shall be cause for rejection of the lot represented by the sample.

3.2 Class 1, style A and style B. The screw extractors shall have left-hand spiral flutes or ridges unless otherwise specified (see 7.2). The flutes shall start at the small end and generate working edges at least to the minimum length of the specified taper. The screw extractors shall be heat treated and conform to a hardness of 52 to 60 on the Rockwell C scale.

3.2.1 Class 1, style A. The type I, class 1, style A screw (stud and pipe) extractors, tapered, spiral flutes or ridges, drill shall be similar to Figure 1 and conform to the dimensions specified in Table I.



Figure 1. Screw (stud and pipe) extractor, class 1, style A, tapered, spiral flutes or ridges, drill.

TABLE I. Screw (stud and pipe) extractor, class 1, style A, tapered, spiral flutes or ridges, drill dimensions and requirements

Extractor	Diameter small end (Inches)	Diameter large end (Inches)	Length of taper (Inches)	Overall length (Inches)	Torsional moment minimum (Inch-pounds)
1	0.054 to 0.070	0.117 to 5/32	1/2 to 7/8	1-13/16 to 2-3/16	23
2	5/64 to 3/32	0.174 to 13/64	11/16 to 1-1/16	2-3/16 to 2-9/16	53
3	7/64 to 9/64	15/64 to 17/64	13/16 to 1-3/16	2-9/16 to 2-15/16	153
4	11/64 to 13/64	5/16 to 11/32	15/16 to 1-5/16	2-13/16 to 3-3/16	630
5	15/64 to 17/64	27/64 to 15/32	1-11/32 to 1-29/32	3-3/16 to 3-9/16	1,050
6	11/32 to 3/8	19/32 to 5/8	1-1/2 to 2	3-9/16 to 3-15/16	3,010
7	7/16 to 1/2	3/4 to 7/8	1-15/16 to 2-5/16	3-15/16 to 4-5/16	6,552
8	23/32 to 3/4	63/64 to 1-3/64	1-15/16 to 2-5/16	4-3/16 to 4-5/8	16,800

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TABLE I. Screw (stud and pipe) extractor, class 1, style A, tapered, spiral flutes or ridges, drill dimensions and requirements (continued)

Extractor	Diameter small end (Inches)	Diameter large end (Inches)	Length of taper (Inches)	Overall length (Inches)	Torsional moment minimum (Inch-pounds)
9	31/32 to 1	1-1/4 to 1-5/16	2-1/16 to 2-7/16	4-7/16 to 5-1/32	30,000
10	1-7/32 to 1-1/4	1-1/2 to 1-9/16	2-5/16 to 2-3/4	4-13/16 to 5-1/4	48,000
11	1-15/32 to 1-1/2	1-13/16 to 1-7/8	2-13/16 to 3-1/4	5-7/16 to 5-7/8	60,000
12	1-27/32 to 1-7/8	2-1/4 to 2-3/8	3-5/16 to 3-11/16	6-1/16 to 6-1/2	60,000

3.2.2 Class 1, style B. The class 1, style B screw (stud and pipe) extractors, tapered, spiral flutes or ridges, plug shall be similar to Figure 2 and conform to the dimensions specified in Table II.

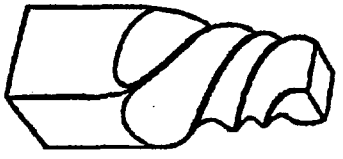


Figure 2. Screw (stud and pipe) extractor, class 1, style B, tapered, spiral flutes or ridges, plug.

TABLE II. Screw (stud and pipe) extractor, class 1, style B, tapered, spiral flutes or ridges, plug dimensions and requirements

Extractor size number	Nominal dimension across corners, small end (Inches)	Nominal dimensions across corners, large end (Inches)	Length of taper minimum (Inches)	Overall length minimum (Inches)	Torsional moment minimum (Inch-pounds)
1	3/16	7/16	3/4	1-1/2	650
2	1/4	17/32	13/16	1-5/8	1,500
3	3/8	45/64	15/16	1-3/4	2,400
4	13/32	7/8	1-1/16	1-15/16	3,000
10	15/32	1-1/16	1-7/16	2-5/8	4,000
5	11/16	1-1/8	1-9/16	2-5/8	6,800
6	7/8	1-9/64	1-13/16	2-13/16	14,000
7	1-5/32	1-15/16	2-3/16	3-3/8	30,000
8	1-13/32	2-1/8	2-5/16	3-1/2	60,000
9	2-1/32	2-53/64	3-1/4	4-1/2	60,000

3.3 Class 2. The class 2 screw (stud and pipe) extractors, tapered, straight fluted, and full length of taper shall be similar to Figure 3 and conform to the dimensions specified in Table III. The screw extractors shall be heat treated and conform to a hardness of 52 to 60 on the Rockwell C scale.



Figure 3. Type I, class 2, screw and pipe extractors, tapered, straight fluted.

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TABLE III. Class 2, screw (stud and pipe) extractors, tapered, straight fluted dimensions and requirements

Extractor size number	Across corners, dimension, small end (Inch)	Across flats, dimension, large end (Inch)	Length of taper ± 1/16 (Inches)	Overall length (Inches)	Torsional moment minimum (Inch-pounds)
1	1/8	3/16	1	2-5/16	90
2	11/64	3/16	7/8	2-5/8	160
3	13/64	1/4	1-11/32	2-7/8	320
4	19/64	5/16	1-7/16	3	500
5	11/32	3/8	1-5/8	3-1/4	670
6	27/64	7/16	1-23/32	3-7/8	900
7	1/2	1/2	2-13/32	4-1/2	1,250
8	19/32	5/8	2-9/16	4-7/8	1,650
9	23/32	3/4	2-15/32	5-3/8	2,100
10	31/32	1	3	5-1/2	3,500

3.4 Class 3. The class 3 screw (stud and pipe) extractors, straight, straight fluted, square tang shall be similar to Figure 4 and conform to the dimensions specified in Table IV for standard pipe, or Table V for extra strong pipe. The screw extractors shall be heat treated and conform to a hardness of 50 to 60 on the Rockwell C scale.



Figure 4. Class 3, screw (stud and pipe) extractors, straight, straight fluted square tang.

TABLE IV. Class 3, screw (stud and pipe) extractors, straight, straight fluted, standard pipe dimensions and requirements

Extractor size number	Nominal standard pipe size (Inch)	Diameter over flutes (Inches)	Body diameter between flutes (Inches)	Length of flutes ± 1/16 (Inches)	Overall length ± 1/16 (Inches)	Torsional moment, minimum (Inch-pounds)	Size drill for torsional test (Inches)
1	1/8	0.286 to 0.290	0.244 to 0.250	3/4	2-1/4	600	1/4
2	1/4	0.387 to 0.391	0.341 to 0.347	27/32	2-1/2	1,500	23/64
3	3/8	0.520 to 0.524	0.466 to 0.472	29/32	2-3/4	3,000	31/64
4	1/2	0.653 to 0.657	0.581 to 0.588	1	3	5,000	39/64
5	3/4	0.858 to 0.862	0.782 to 0.790	1	3-1/4	10,000	13/16
6	1	1.092 to 1.096	1.008 to 1.016	1	3-1/2	20,000	1-1/32
7	1-1/4	1.433 to 1.437	1.332 to 1.340	1-1/4	3-3/4	25,000	1-23/64
8	1-1/2	1.672 to 1.676	1.562 to 1.570	1-3/8	4	30,000	1-19/32
9	2	2.141 to 2.146	2.014 to 2.022	1-1/2	4-1/2	40,000	2-3/64

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TABLE IV. Class 3, screw (stud and pipe) extractors, straight, straight fluted, extra strong pipe dimensions and requirements

Extractor size number	Nominal extra strong pipe size (Inch)	Diameter over flutes (Inches)	Body diameter between flutes (Inches)	Length of flutes ± 1/16 (Inches)	Overall length ± 1/16 (Inches)	Torsional moment, minimum (Inch-pounds)	Size drill for torsional test (Inches)
10	1/8	0.225 to 0.229	0.183 to 0.189	3/4	2-1/4	220	13/64
11	1/4	0.318 to 0.322	0.272 to 0.278	27/32	2-1/2	700	19/64
12	3/8	0.443 to 0.447	0.389 to 0.396	29/32	2-3/4	1700	13/32
13	1/2	0.569 to 0.573	0.499 to 0.507	1	3	5000	17/32
14	3/4	0.769 to 0.773	0.685 to 0.693	1	3-1/4	10,000	23/32
15	1	0.994 to 0.998	0.910 to 0.916	1	3-1/2	20,000	15/16
16	1-1/4	1.325 to 1.329	1.223 to 1.231	1-1/4	3-3/4	25,000	1-17/64
17	1-1/2	1.556 to 1.560	1.445 to 1.453	1-3/8	4	30,000	1-31/64
18	2	2.007 to 2.012	1.879 to 1.887	1-1/2	4-1/2	40,000	1-15/16

3.5 Identification marking. The tool shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.6 Workmanship. Details of workmanship shall be in accordance with the best commercial practice. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g., sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

3.7 Extractor sets. When sets of extractors are specified (see 7.2), the parts of the sets shall conform to the requirements of the respective items covered by this commercial item description. The sets shall be provided in wraps or rolls, be of durable plastic or fabric with individual pockets for each extractor and its components, arranged to fold in a manner to enclose the extractor and its components, and shall be secured with durable snaps or ties. When boxes for sets are specified, they shall be either metal, plastic, or wood with partitions for holding the components and a cover, having a positive- or friction-type catch. When wooden blocks are specified, they shall have round drilled holes to accommodate each extractor in the set. The interior and exterior surfaces of metal, wooden boxes, or wooden blocks shall have a suitable protective coating. When plastic wraps (pouches) are furnished, the plastic shall be of flexible vinyl film having a 1.58 maximum volatility rating, 1.30 maximum specific gravity, 13 gage (0.013 inch) minimum thickness and shall have the capacity to pass a 0° to -5° Fahrenheit Maslan cold crack test.

4. REGULATORY REQUIREMENTS.

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4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The product provided to the Government shall conform to the contractually specified requirements. For product characteristics and requirements which are not contractually specified, the provided product shall conform to the producer's own drawings, standards, specifications, and quality assurance practices for the highest quality product commercially offered by the producer. If the product is not commercially offered by the producer, the best commercial practices shall apply. The Government reserves the right to require proof of such conformances prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to assure the item conforms to the specified requirements.

5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics by the Government. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING.

6.1 Packaging. Requirements of preservation, packing, packaging, and marking of packages shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of referenced documents:

Federal Specifications and Standards: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC 20407.

Federal Acquisition Regulations (FAR): Government Printing Office, Superintendent of Documents, Washington, DC 20402-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 43201-4606.

ASTM E18: American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

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7.2 Ordering data. Purchasers should select the preferred options permitted herein and should include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) When applicable, the appropriate NSN(s).
(c) When applicable, the specific class and style, as specified in the Commercial Item Description.
(d) If bid samples are required, the number of samples required, and the subjective inspection requirements.
(e) If first article samples are required and the number of samples required.
(f) If lot sampling inspection is required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Inspection. Each sample tool shall be examined and tested for all salient characteristics. The Inspection Level shall be S-2 with an Acceptable Quality Level of 4.0, expressed in terms of defects per hundred units.

(g) What preservation, packing, packaging, and marking of packages are required.
(h) If other than left-hand spiral flutes or ridges are required (see 3.2).
(i) Whether sets are to be furnished and type of wrap or container; i.e., plastic or fabric wraps or rolls, or metal, plastic, wooden boxes, or wooden blocks (see 3.7).

7.3 National Stock Numbers (NSNs). The following list of NSNs corresponds to the classes and styles specified in this CID. This list may not be indicative of all possible NSNs associated with this CID.

Extractor (spiral, fluted, and tapered)		**Extractor (straight, fluted, and tapered)**	
NSN	SIZE	NSN	SIZE
5120-00-240-5223	1	5120-00-180-0777	1
5120-00-580-2359	2	5120-00-180-0778	2
5120-00-240-5221	3	5120-00-180-0779	3
5120-00-240-5222	4	5120-00-180-0780	4
5120-00-240-5219	5		
5120-00-240-5220	6		
5120-00-240-5217	7		
5120-00-242-1118	8		
5120-00-240-5215	9		
5120-00-223-6954	11		
5120-00-223-6955	12		

MILITARY INTERESTS:

NONE: DoD has no registered interest in revisions or notices to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

GSA - FSS

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